Yutao Li

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PROFESSIONAL SUMMARY

Computer Engineering M.S. student at UC San Diego with experience in machine learning, computer vision, LLM agents, and robotics. Published a first-author paper at IEEE VTC and developed PyTorch models, AI agent systems, and SLAM pipelines.

EDUCATION

University of California, San Diego (UCSD) — San Diego, CA Sep. 2025 – Dec. 2026
M.S. in Electrical and Computer Engineering — Computer Engineering
Cumulative GPA: **3.93**/4.00

The Chinese University of Hong Kong, Shenzhen (CUHKSZ) — Shenzhen, China Sep. 2021 – Jul. 2025
Bachelor of Engineering in Electronic Information Engineering — Electrical Engineering Track
Cumulative GPA: **3.56**/ 4.00 | Major GPA: **3.63**/4.00

PUBLICATIONS

[1] Y. Li, Y. Wang, and C. Huang, "NeRA: Neural Reflectance and Attenuation Fields for Radio Map Reconstruction," *IEEE Vehicular Technology Conference (VTC)*, 2024. [Paper]
[2] Y. Wang, **Y. Li**, G. Chen, and C. Huang, "Channel Knowledge Maps Construction Based on Point Cloud Environment Information," *IEEE Vehicular Technology Conference (VTC)*, 2024. [Paper]

RESEARCH

NeRA: Neural Reflectance and Attenuation Fields for Radio Map Reconstruction Jul. 2022 – May 2024
Research Assistant; Supervised by Prof. Chuan Huang, Future Network of Intelligence Institute, CUHKSZ

- Developed PyTorch-based neural reflectance and attenuation fields that modeled real-world objects as learnable media interacting with radio-frequency signals for environment-aware radio map reconstruction.
- Designed and implemented the end-to-end research pipeline, covering environmental data preprocessing, neural network architecture development, model training, experiment configuration, quantitative evaluation, and result visualization.
- Evaluated reconstruction accuracy, robustness, and cross-environment generalization against baseline methods across multiple real-world layouts. Reduced mean absolute prediction error by 26% relative to NeRF^2.
- Led model implementation, real-world empirical data collection, experimental debugging, result analysis, figure generation, and manuscript preparation, resulting in a first-author publication at IEEE VTC 2024.

Remote Sensing Semantic Segmentation Based on Multimodal Multi-Scale LoRA-Tuned SAM Sep. 2024 – Dec. 2024
Research Assistant; Supervised by Prof. Simon Pun, School of Science and Engineering, CUHKSZ

- Implemented a multimodal remote-sensing semantic-segmentation framework by adapting the Segment Anything Model (SAM) with multiple Low-Rank Adaptation (LoRA) modules configured at different ranks.
- Combined information from multiple remote-sensing inputs through a dedicated feature-fusion stage.
- Implemented a PyTorch-based training and evaluation pipeline covering multimodal data preprocessing, model fine-tuning, segmentation-map generation, and comparative performance analysis across different remote-sensing inputs.
- Investigated Transformer image encoders, attention mechanisms, and parameter-efficient adaptation strategies to improve task-specific feature extraction and segmentation performance for multimodal remote-sensing imagery.

SELECTED PROJECTS

Agentic AI Job Application Assistant [GitHub] [Video Demo] Jun. 2026 – Present
Python, Pydantic, LLM APIs, Streamlit, RAG, ReAct, Docker

- Built an end-to-end AI job application assistant that parses job descriptions into structured data, matches role requirements against candidate profiles, generates tailored application materials, and tracks application progress.
- Integrated a configurable OpenAI-compatible LLM client with environment-based API and model settings, structured response generation, and validation mechanisms for detecting incomplete, malformed, or schema-invalid model outputs.
- Dockerized the product and made the shared backend pipeline compatible with both CLI and Streamlit interface, enabling consistent workflows and reproducible local execution across development environments.
- Built an interactive user interface to present parsed job requirements, profile-match analysis results, and generated application material, allowing users to review each stage of the end-to-end workflow and ask follow-up questions.
- Structured the system as a modular, Pydantic-validated workflow with persistent JSON/Markdown artifacts, making intermediate outputs traceable and reusable across applications.

EDAgent: Agentic LLM Framework for Economic Dispatch Jan. 2026 – Mar. 2026

Python, ReAct, CVXPY, OSQP, FastAPI, Streamlit

- Developed a ReAct-based LLM agent that interpreted natural-language power-system requests, selected domain-specific tools, solved strictly constrained economic-dispatch problems, and visualized optimization results.
- Integrated IEEE 14-, 30-, 57-, and 118-bus cases with support for user-specified generator constraints, using CVXPY/OSQP quadratic programming to minimize generation cost under strict physical constraints.
- Streamlit and FastAPI interfaces let users interact with the agent in natural language to modify grid parameters, run dispatch scenarios, and inspect solver results without directly accessing the optimization backend.
- Improved agent reliability through working memory, post-solver validation, and exception handling for failed outputs.
- Evaluated EDAgent against a prompted LLM baseline across 25 scenarios and five IEEE systems, achieving 88% end-to-end success, 100% feasibility among completed runs, and 0.44% mean cost error versus 56% success and 5.82% error.

Visual-Inertial SLAM Feb. 2026 – Mar. 2026

Python, NumPy, SciPy, OpenCV, EKF, SE(3), Stereo Vision

- Implemented an EKF-based visual-inertial SLAM pipeline that fused IMU linear and angular velocity measurements with calibrated stereo-camera feature observations to jointly estimate the robot's poses and 3D landmark positions.
- Derived and implemented SE(3)-based motion prediction, state-transition Jacobians, and covariance propagation for IMU-driven pose estimation, while tuning process and observation noise parameters to limit accumulated drift.
- Initialized landmarks from stereo correspondences and developed EKF measurement updates using camera intrinsic and extrinsic calibration, feature filtering, and batched matrix operations.
- Evaluated the system on Clearpath Jackal datasets collected on the MIT campus, visualizing estimated trajectories and landmark maps and analyzing errors caused by sensor noise and limited vertical motion.
- Across two Clearpath Jackal sequences, stereo updates corrected accumulated IMU drift and produced more physically consistent trajectories and landmark maps than IMU-only localization.

Quantized CNN Deployment on a Verilog Systolic-Array Accelerator Nov. 2025 – Dec. 2025

Python, PyTorch, Verilog, Quantization, Systolic Array, RTL Verification

- Built an end-to-end model-to-RTL deployment workflow that quantized a VGG network in PyTorch, selected a convolution layer for hardware execution, and mapped its computations onto an 8×8 Verilog systolic array.
- Translated PyTorch activations and weights into hardware-compatible fixed-point stimulus files and organized the data according to the accelerator's weight-stationary execution schedule.
- Implemented the Verilog inference datapath integrating MAC processing elements, on-chip activation and weight storage, FIFO-based data movement, partial-sum accumulation, and ReLU processing for the mapped CNN layer.
- Developed an RTL testbench that loaded model parameters and activations, executed the mapped convolution, and compared hardware outputs against PyTorch-generated golden results to verify model-to-hardware numerical consistency.
- All hardware simulation outputs matched the PyTorch reference, confirming numerical consistency across the entire flow.

Aircraft Engine Remaining Useful Life Prediction [GitHub] Apr. 2026 – Jun. 2026

Python, PyTorch, LSTM, Time-Series Regression, NASA C-MAPSS

- Reproduced a preprocessing-enhanced LSTM framework for aircraft-engine remaining useful life prediction, implementing the complete training and evaluation workflow across all four NASA C-MAPSS subsets.
- Built a reusable preprocessing pipeline that combined Peason-correlation-based sensor selection with median denoising, Z-score normalization, sliding-window generation, and adaptive piecewise-linear RUL labeling.
- Tuned learning rates and thresholds to align the performance with the reference paper, achieving RMSE values of 8.28, 18.61, 7.04, and 17.43 on FD001–FD004, closely matching the results reported in the reference paper.
- The resulting data pipeline became the shared foundation for the team's recurrent, graph, and attention-based experiments.

TECHNICAL SKILLS

- **Programming Languages**: Python, C++, MATLAB, Verilog
- **ML / AI**: PyTorch, Transformers, LoRA, LLM Agents, RAG, Computer Vision, Multimodal Learning
- **Robotics**: SLAM, EKF, Sensor Fusion, SE(3), GTSAM
- **Frameworks / Libraries**: NumPy, pandas, OpenCV, scikit-learn, CVXPY, Pydantic, FastAPI, Streamlit
- **Tools:** Docker, Git, Linux, ChromaDB

HONORS & AWARDS

- Bowen Scholarship, Chinese University of Hong Kong, Shenzhen Sep. 2021- Jul. 2025
- School of Science and Engineering Dean's List (Twice) 2023, 2024